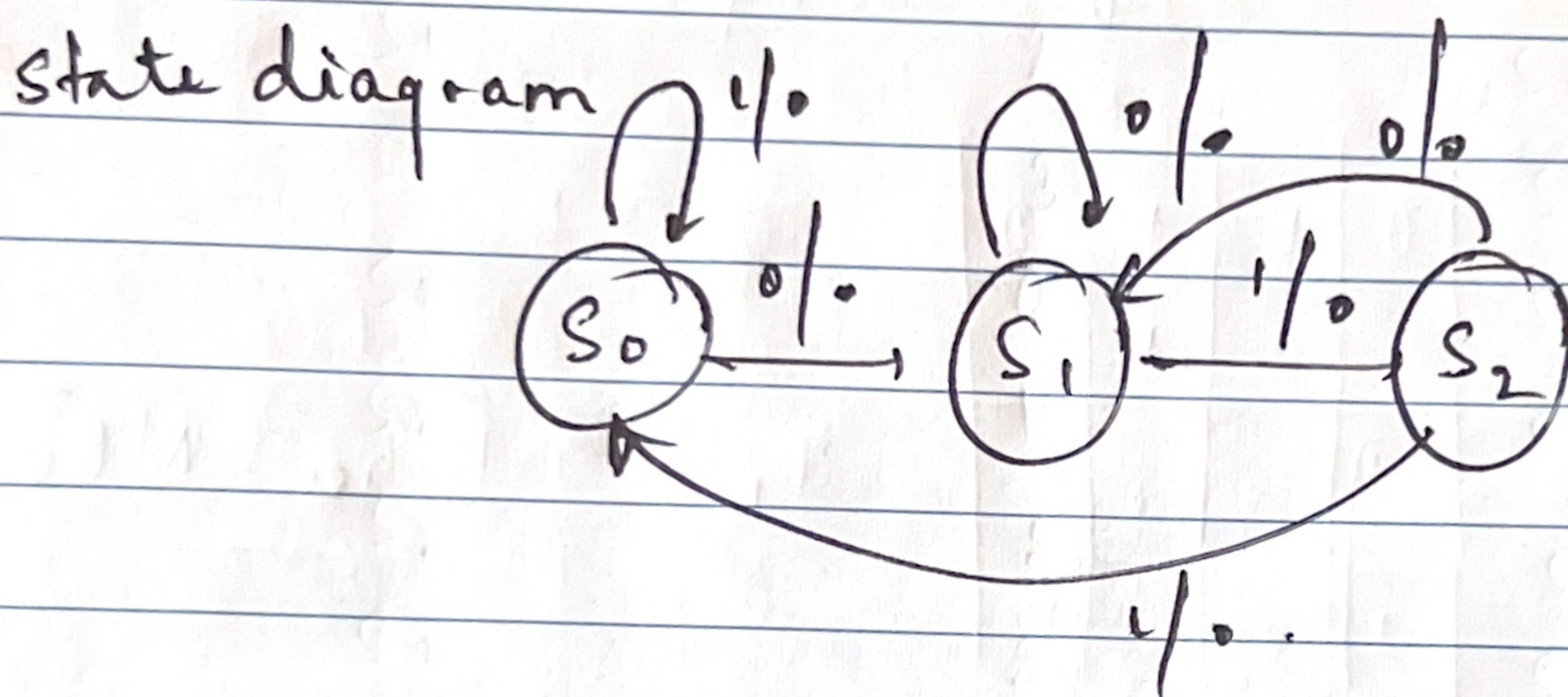


Day 9

FSM Assignment

1) Draw the state diagram for a FSM, the output toggles whenever it detects '110', Consider initial value of output as 0



Previous state	I/P		next state	O/P y.
Q ₁ Q ₀	I		D ₁ D ₀	y
0 0	0		0 1	0
0 0	1		0 0	0
0 1	0		0 1	0
0 1	1		1 0	0
1 0	0		0 1	0
1 0	1		0 0	1
x x	x		x x	0
x x	x		x x	0

		$q_1 \backslash q_0 I$			
		00	01	10	11
$D_1 =$	0	0	0	1	0
	1	0	0	x	x

		$q_1 \backslash q_0 I$			
		00	01	11	10
$D_0 =$	0	1	0	0	1
	1	1	0	x	x

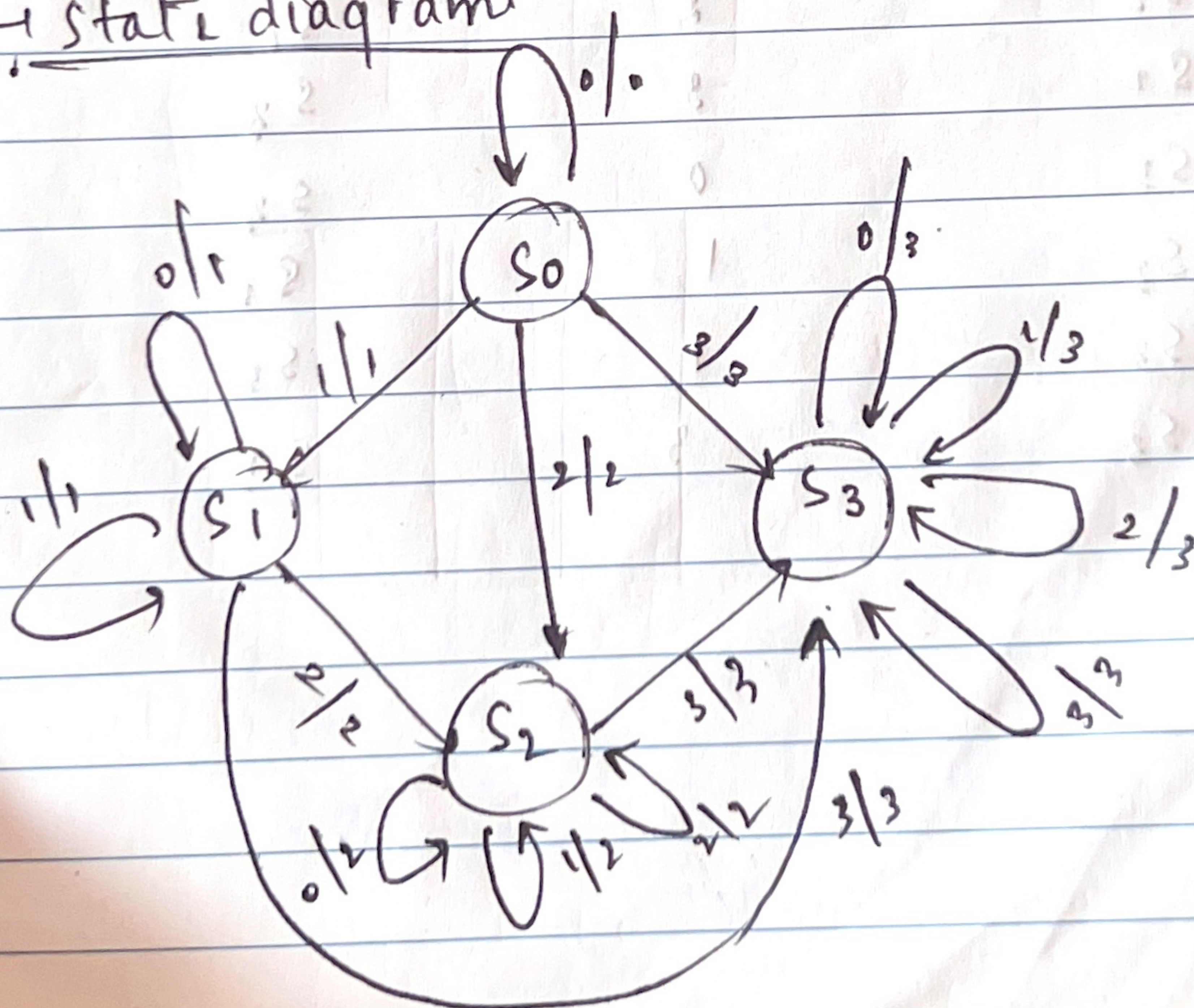
$$D_1 \Rightarrow q_0 I$$

$$D_0 = \bar{I}$$

$$y = q_1 \bar{q}_0 I$$

2) Design a FSM over $\{0, 1, 2, 3\}$

→ state diagram



Previous state

Input

next state

 S_0

0

 S_0 S_0

1

 S_1 S_0

2

 S_2 S_1

3

 S_3 S_1

0

 S_0 S_1

1

 S_1 S_1

2

 S_2 S_1

3

 S_3 S_2

0

 S_0 S_2

1

 S_1 S_2

2

 S_2 S_2

3

 S_3 S_3

0

 S_0 S_3

1

 S_1 S_3

2

 S_2 S_3

3

 S_3